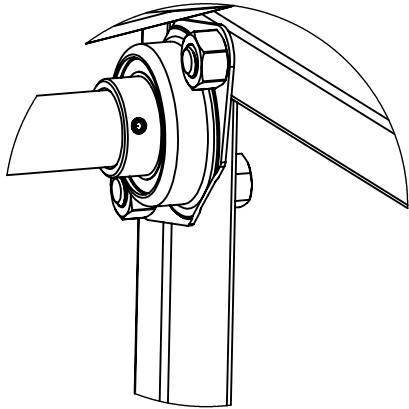
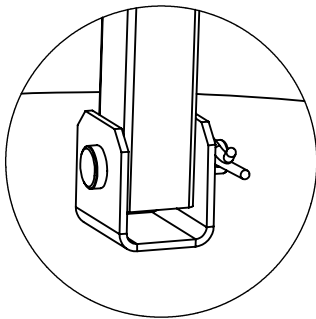


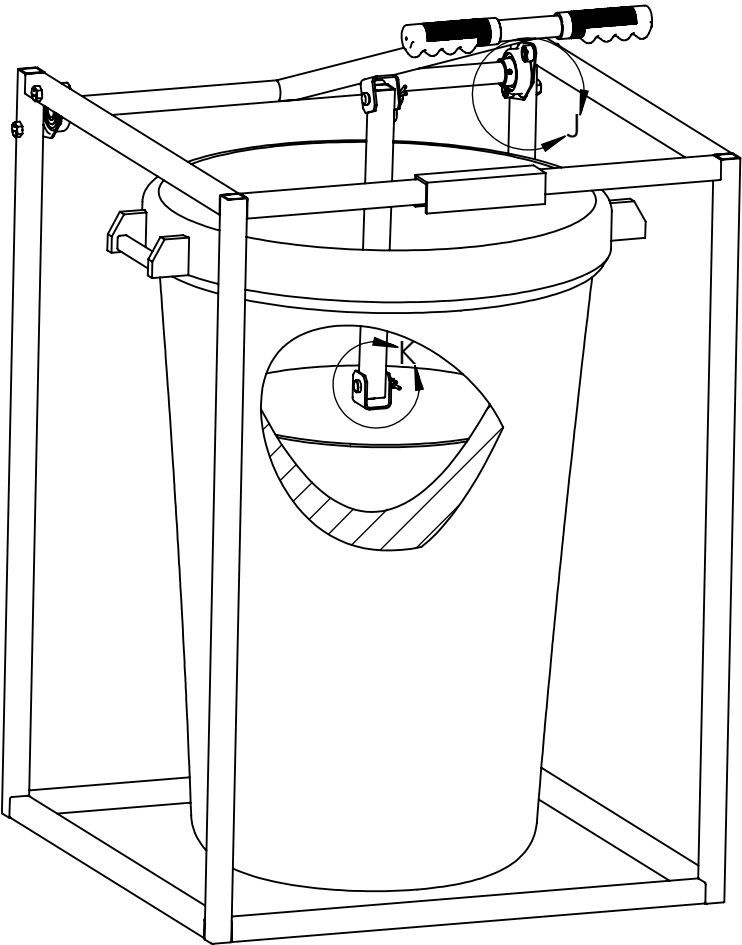
REVISIONS			
REV.1	DESCRIPTION	DATE	APPROVED
1	ADDED TRASHCAN FOR CLARITY	11/18/2025	ID



DETAIL J



DETAIL K



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX ±.01 .XXX ±.005
FRACTIONS ±1/64
ANGULAR MACHINED ±.5°
ANGULAR BEND ±1°
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

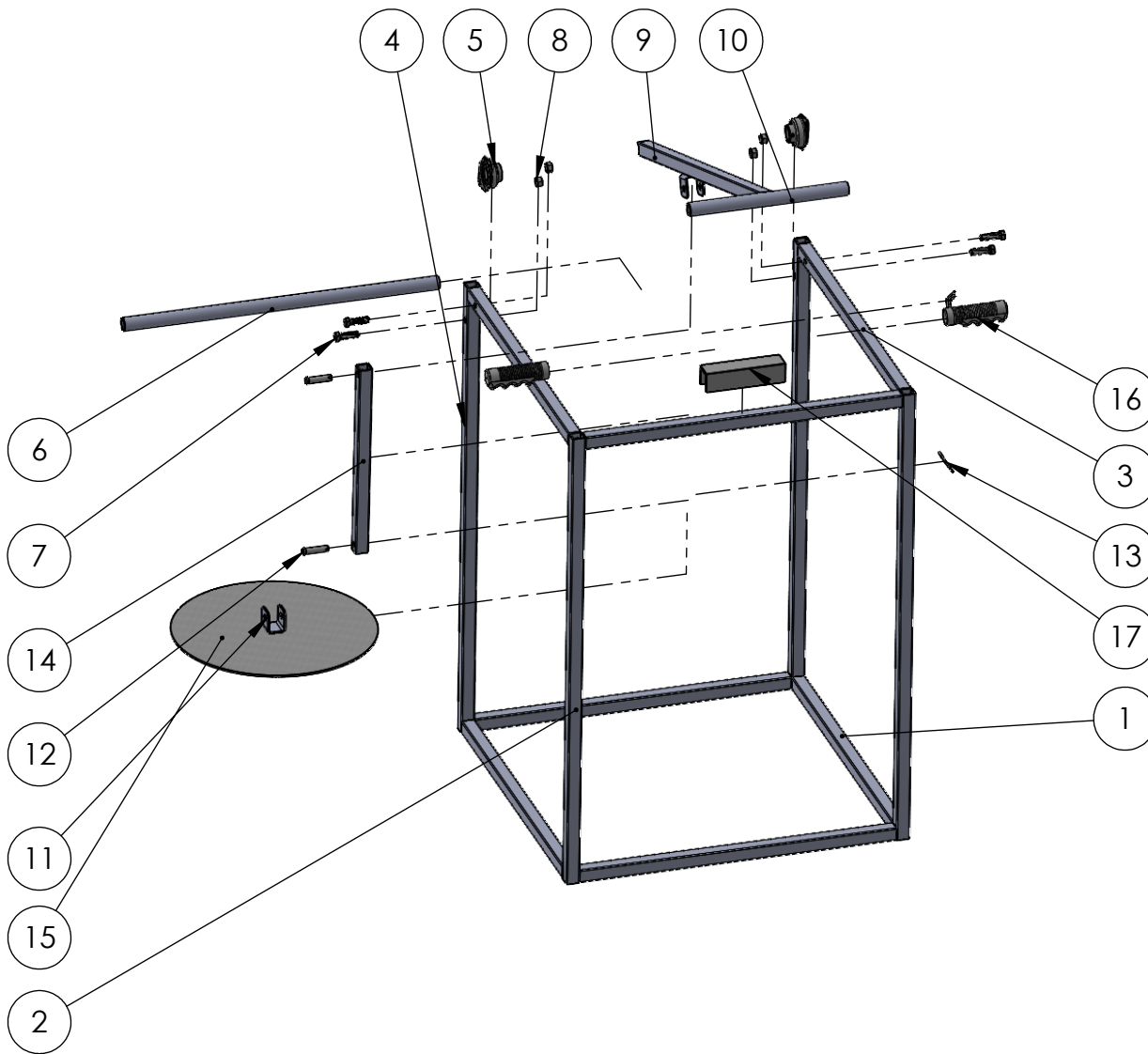
SHEET 1 OF 2



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE: MANUAL TRASH COMPACTOR			
SIZE: A	DATE: 10/29/2025	DRAWN BY: ID	
MATERIAL: VARIED	PART #: A-160GI-002	REV: 0	



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	P-160GI-007	HORIZONTAL TUBE	5
2	P-160GI-006	VERTICAL TUBE	2
3	M-160GI-006	HORIZONTAL TUBE WITH HOLE	2
4	M-160GI-008	VERTICAL TUBE WITH HOLE	2
5	P-160GI-004	BALL BEARING BLOCK	2
6	M-160GI-004	PIVOT PIPE	1
7	P-160GI-001	3/8 HEX BOLT	4
8	P-160GI-002	3/8 HEX NUT	4
9	M-160GI-007	HANDLE SHAFT	1
10	M-160GI-003	HANDLE	1
11	M-160GI-002	PIVOT BRACKET	2
12	P-160GI-003	CLEVIS PIN	2
13	P-160GI-005	LOCKING PIN	2
14	M-160GI-005	COMPRESS SHAFT	1
15	M-160GI-001	COMPRESS PLATE	1
16	P-160GI-008	HAND GRIPS	2
17	P-160GI-009	RUBBER PAD	1

REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
1	ADDED HAND GRIPS AND PADS	11/17/25	ID

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

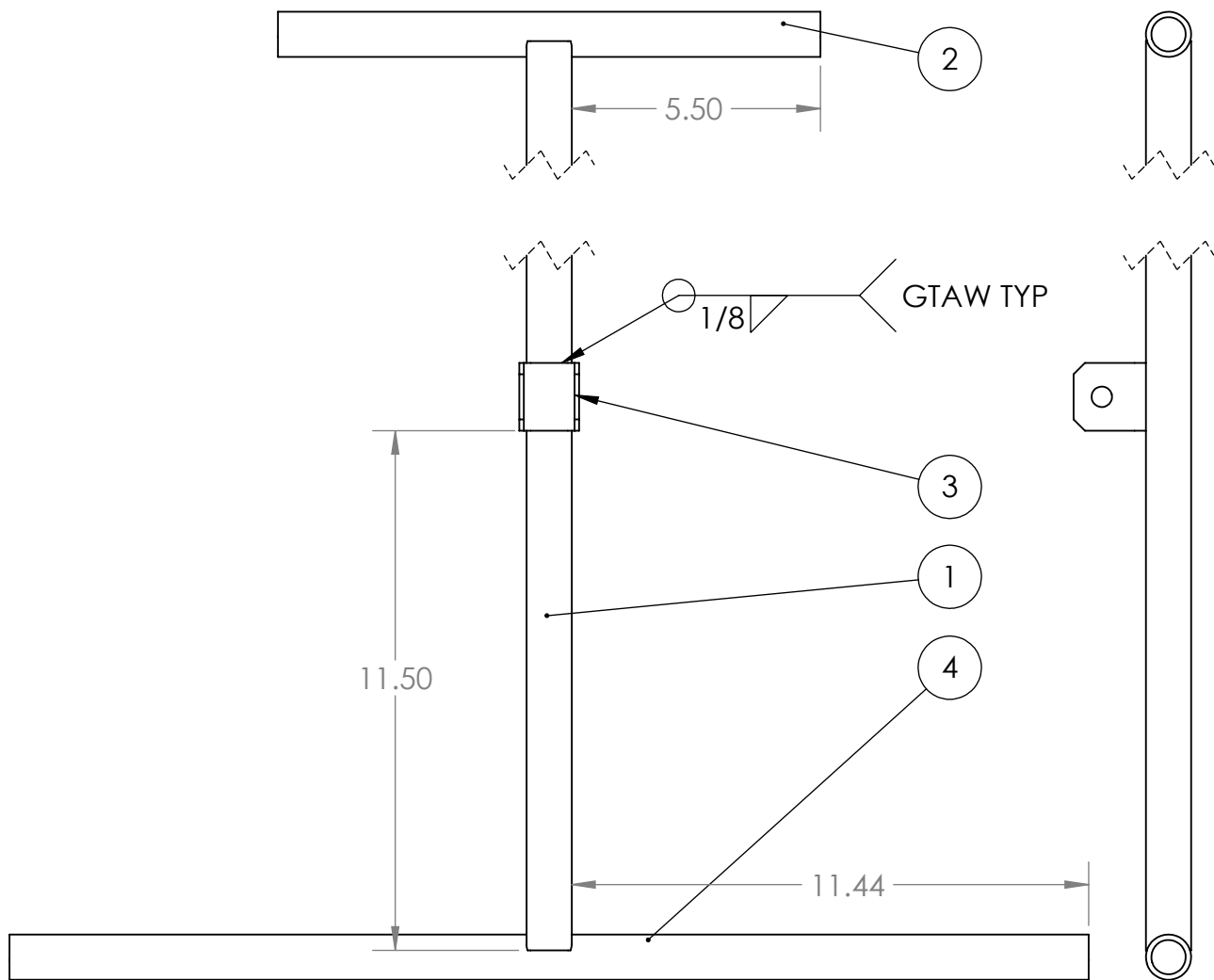
TOLERANCES:
 .XX $\pm$.01 .XXX $\pm$.005
 FRACTIONS $\pm$ 1/64
 ANGULAR MACHINED $\pm$.5°
 ANGULAR BEND $\pm$ 1°
 SURFACE FINISH
 BREAK ALL EDGES .030

DO NOT SCALE
SHEET 2 OF 2



CASE SCHOOL
OF ENGINEERING
CASE WESTERN RESERVE
UNIVERSITY

TITLE: MANUAL TRASH COMPACTOR			
SIZE: A	DATE: 10/29/2025	DRAWN BY: ID	
MATERIAL: VARIED	PART #: A-160GI-002	REV: 1	



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	M-160GI-007	HANDLE SHAFT	1
2	M-160GI-003	HANDLE	1
3	M-160GI-002	PIVOT BRACKET	1
4	M-160GI-004	PIVOT PIPE	1
REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
1	CORRECTED WELD SYMBOLS	11/18/2025	ID

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm$.01 .XXX $\pm$.005
FRACTIONS $\pm$ 1/64
ANGULAR MACHINED $\pm$.5°
ANGULAR BEND $\pm$ 1°
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET 1 OF 1



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

HANDLE SHAFT SUB ASSEMBLY

SIZE:

A

DATE:

11/18/2025

DRAWN BY:

ID

MATERIAL:

LOW CARBON STEEL

PART #:

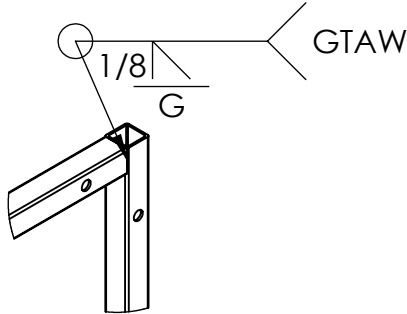
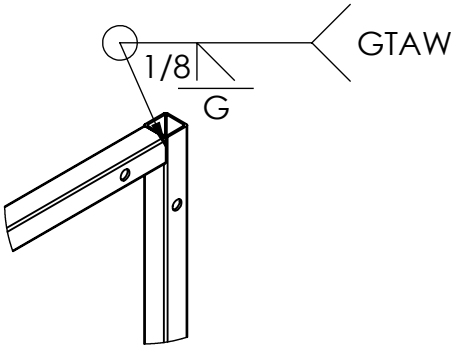
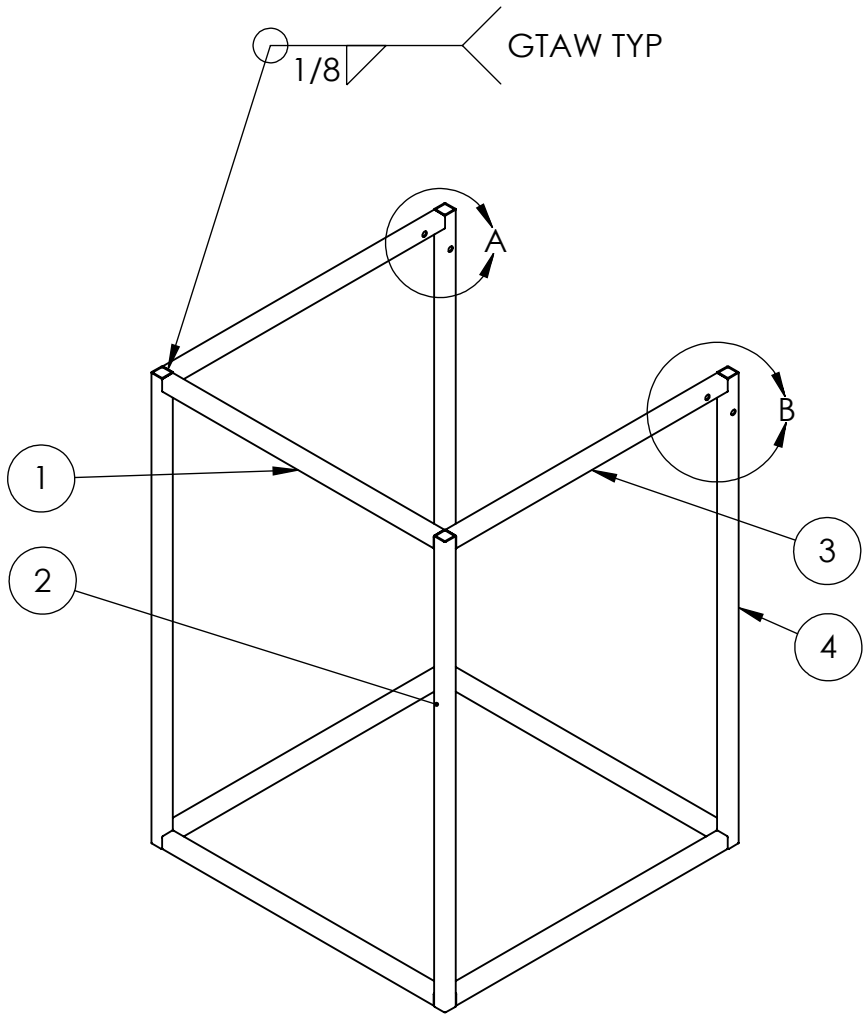
SA-160-001

REV:

1

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	P-160GI-007	HORIZONTAL TUBE	5
2	P-160GI-006	VERTICAL TUBE	2
3	M-160GI-006	HORIZONTAL TUBE WITH HOLE	2
4	M-160GI-008	VERTICAL TUBE WITH HOLE	2

REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
1	MODIFIED WELDING SYMBOLS	11/18/2025	ID



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX ±.01 .XXX ±.005
FRACTIONS ±1/64
ANGULAR MACHINED ±.5°
ANGULAR BEND ±1°
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET 1 OF 1



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

MAIN FRAME

SIZE:

A

DATE:

11/18/2025

DRAWN BY:

ID

MATERIAL:

LOW CARBON STEEL

PART #:

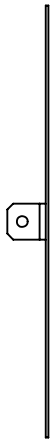
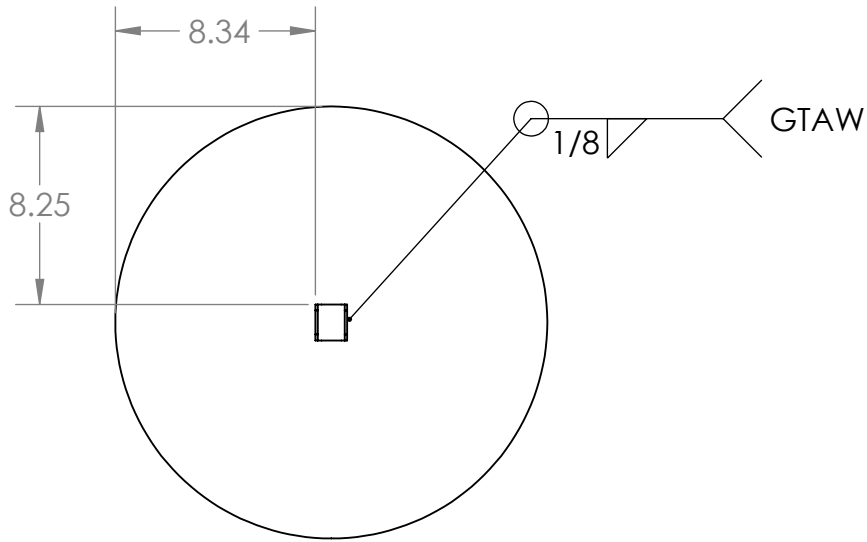
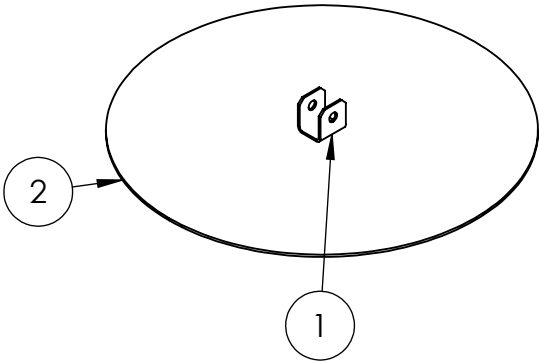
SA-160GI-002

REV:

REV

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	M-160GI-002	PIVOT BRACKET	1
2	M-160GI-001	COMPRESS PLATE	1

REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
1	REMOVED DETAIL VIEW	11/18/25	NTA25
2	ADDED SIDE VIEW	11/18/25	NTA25
3	FIXED WELDING TYPE	11/18/25	NTA25



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm$.01 .XXX $\pm$.005
FRACTIONS $\pm$ 1/64
ANGULAR MACHINED $\pm$.5°
ANGULAR BEND $\pm$ 1°
SURFACE FINISH
BREAK ALL EDGES .030

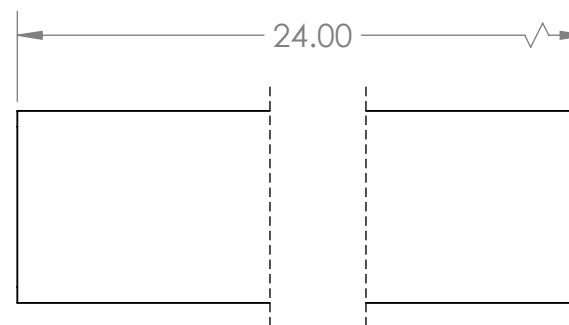
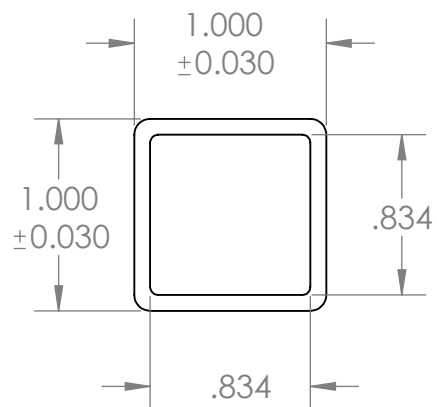


CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE: PLATE SUB ASSEMBLY			
SIZE: A	DATE: 10/29/2025	DRAWN BY: NTA25	
MATERIAL: VARIED	PART #: SA-160GI-003	REV: 1	

DO NOT SCALE
SHEET 1 OF 1



McMASTER CARR
PART NUMBER
6527K262

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX ±.01 .XXX ±.005
FRACTIONS ±1/64
ANGULAR MACHINED ±.5°
ANGULAR BEND ±1°
SURFACE FINISH
BREAK ALL EDGES .030



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

Horizontal Tube No Hole

SIZE:

A

DATE:

10/26/2025

DRAWN BY:

LC

MATERIAL:

Low-Carbon Steel

PART #:

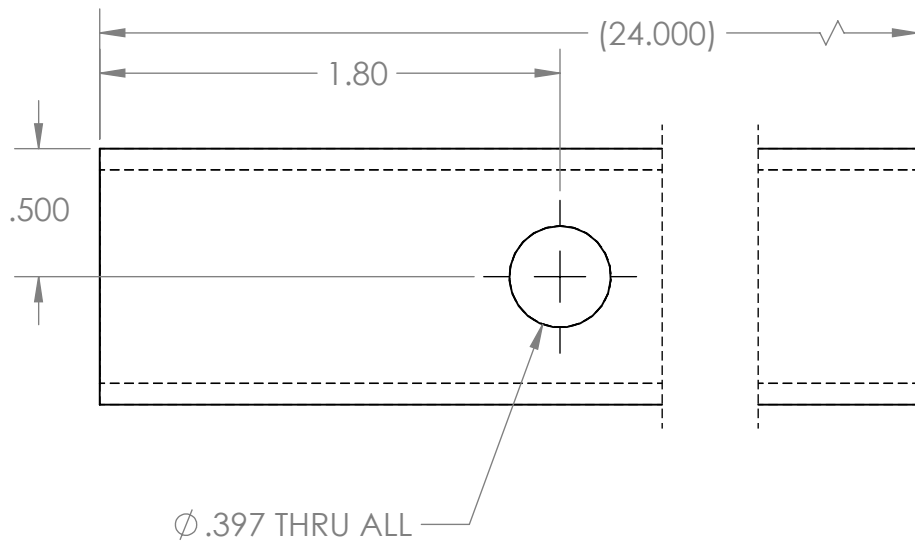
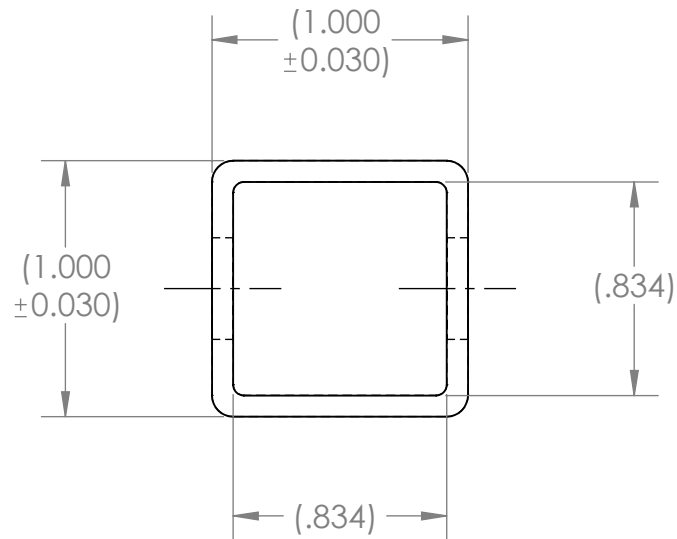
P-160GI-007

REV:

1

DO NOT SCALE

SHEET 1 OF 1



NOTE: TO BE MANUFACTURED
 BY MODIFYING McMASTER CARR
 PART NUMBER 6527K262

UNLESS OTHERWISE SPECIFIED:
 DIMENSIONS IN INCHES

TOLERANCES:
 .XX $\pm .01$.XXX $\pm .005$
 FRACTIONS $\pm 1/64$
 ANGULAR MACHINED $\pm .5^\circ$
 ANGULAR BEND $\pm 1^\circ$
 SURFACE FINISH
 BREAK ALL EDGES .030



CASE SCHOOL
 OF ENGINEERING

CASE WESTERN RESERVE
 UNIVERSITY

TITLE:

Horizontal Tube With Hole

SIZE:

A

DATE:

10/27/2025

DRAWN BY:

LC

MATERIAL:

Low-Carbon Steel

PART #:

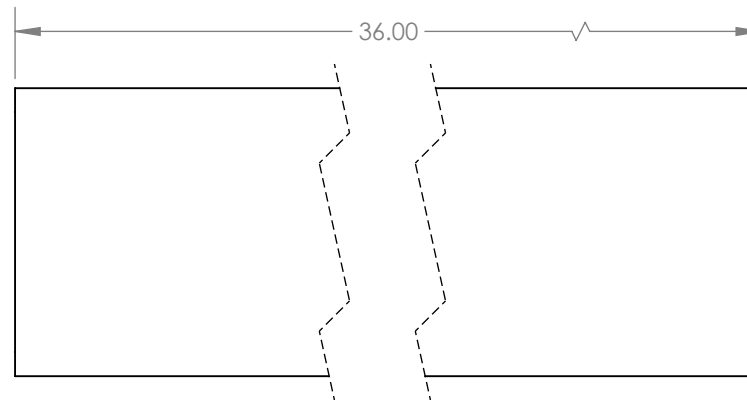
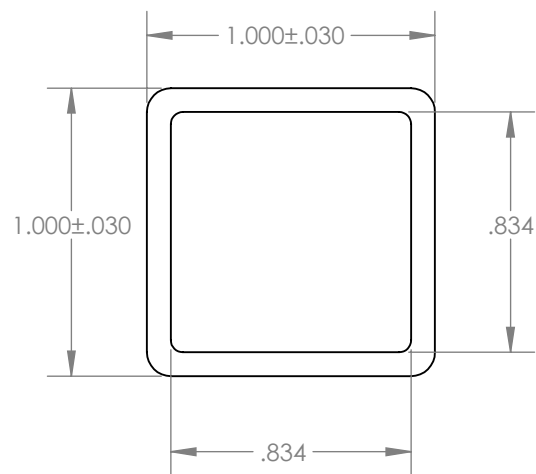
M-160GI-007

REV:

1

DO NOT SCALE

SHEET 1 OF 1



McMASTER CARR
PART NUMBER
6527K262

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX ±.01 .XXX ±.005
FRACTIONS ±1/64
ANGULAR MACHINED ±.5°
ANGULAR BEND ±1°
SURFACE FINISH
BREAK ALL EDGES .030



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

VERTICAL BEAM

SIZE:

A

DATE:

10/28/2025

DRAWN BY:

AXH1066

MATERIAL:

Low-Carbon Steel

PART #:

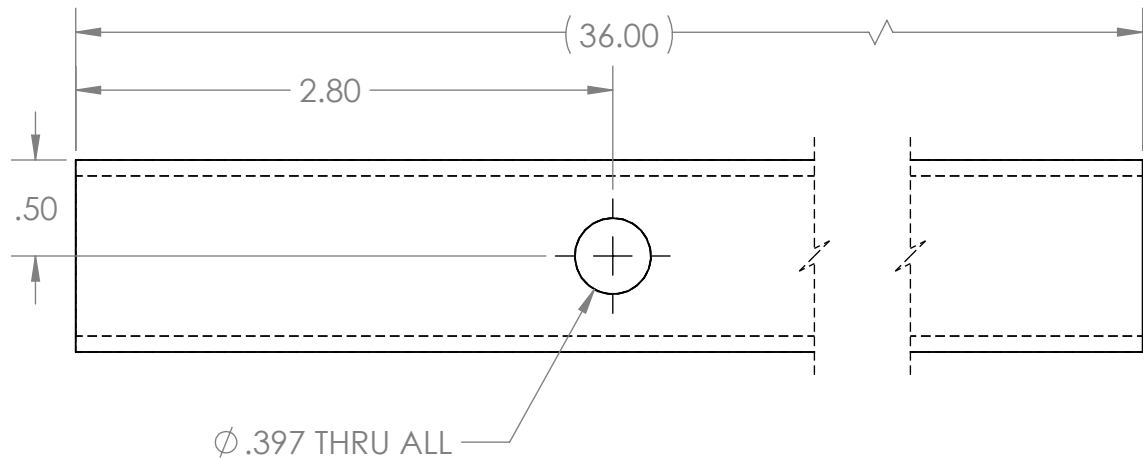
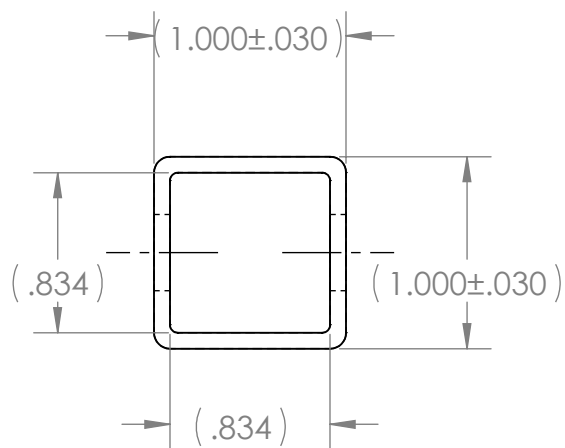
P-160GI-006

REV:

0

DO NOT SCALE

SHEET 1 OF 1



Manufactured From
McMaster-Carr Part
6527K263

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm .01$.XXX $\pm .005$
FRACTIONS $\pm 1/64$
ANGULAR MACHINED $\pm .5^\circ$
ANGULAR BEND $\pm 1^\circ$
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET 1 OF 1



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

VERTICAL TUBE WITH HOLE

SIZE:

A

DATE:

10/28/2025

DRAWN BY:

AXH1066

MATERIAL:

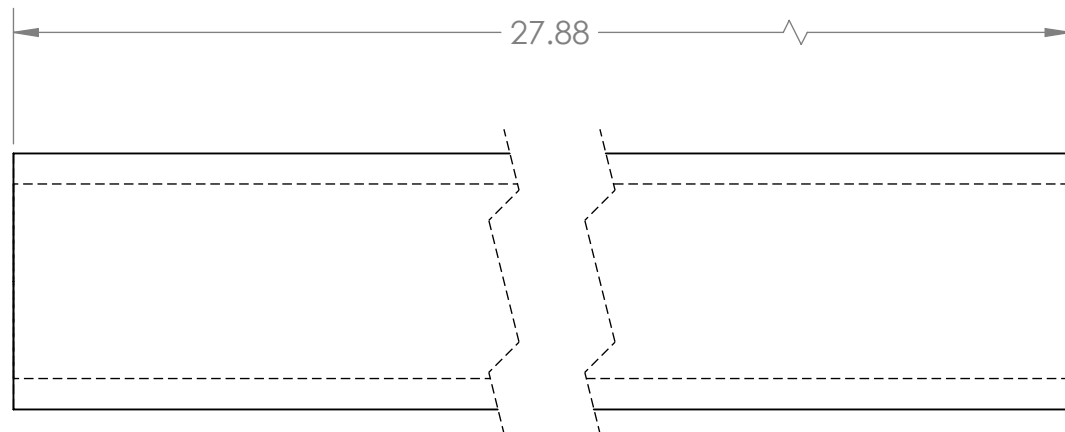
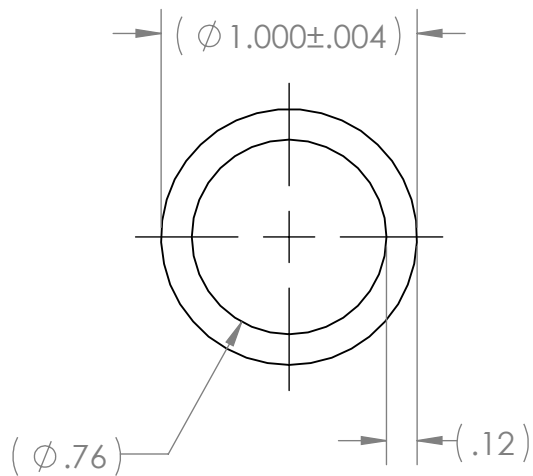
LOW CARBON STEEL

PART #:

M-160GI-008

REV:

0



Manufactured From
 McMaster-Carr Part
 7767T372

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
 .XX ± .01 .XXX ± .005
 FRACTIONS ± 1/64
 ANGULAR MACHINED ± .5°
 ANGULAR BEND ± 1°
 SURFACE FINISH
 BREAK ALL EDGES .030



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

ROTATING SHAFT

SIZE:

A

DATE:

10/27/2025

DRAWN BY:

AXH1066

MATERIAL:

Low-Carbon Steel

PART #:

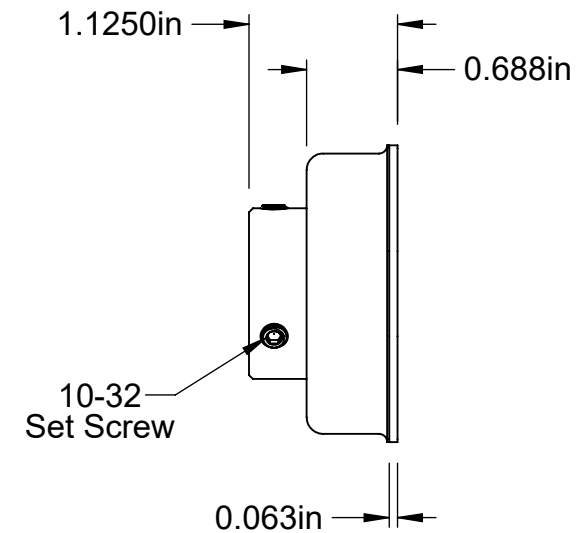
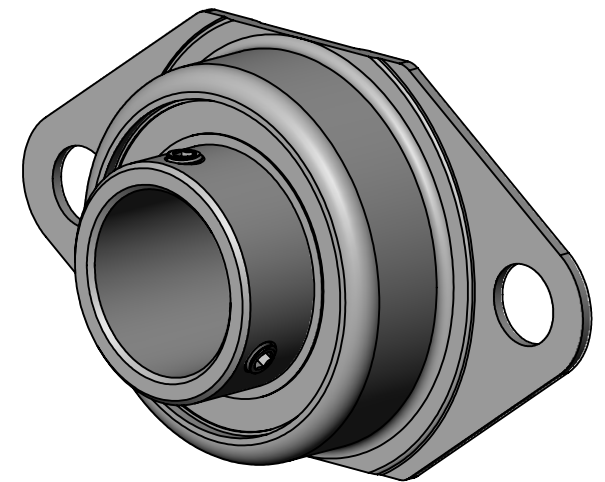
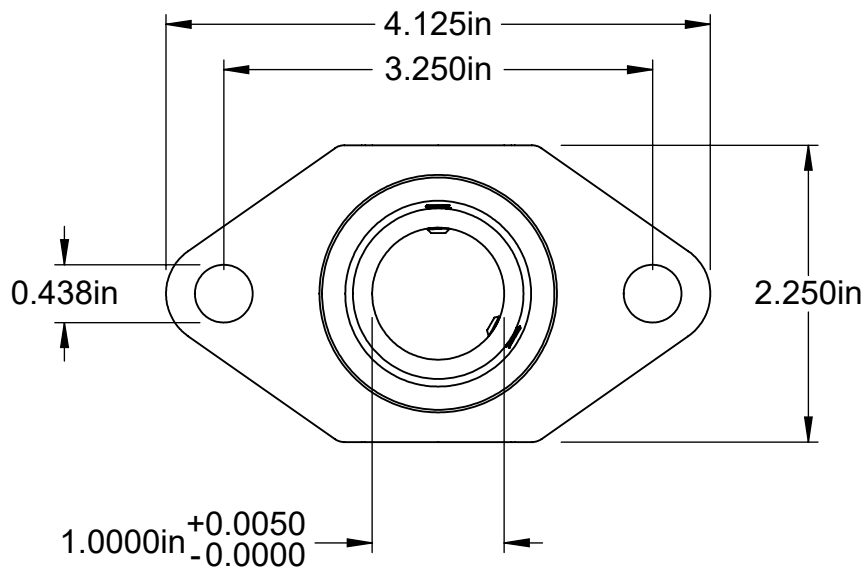
M-160GI-004

REV:

0

DO NOT SCALE

SHEET 1 OF 1



MCMASTER CARR
PART NUMBER
7208K55

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm$.01 .XXX $\pm$.005
FRACTIONS $\pm$ 1/64
ANGULAR MACHINED $\pm$.5°
ANGULAR BEND $\pm$ 1°
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE
SHEET1 OF 1



**CASE SCHOOL
OF ENGINEERING**

**CASE WESTERN RESERVE
UNIVERSITY**

TITLE:

BALL BEARINGS

SIZE:

A

DATE:

8/23/2010

DRAWN BY:

NTA25

MATERIAL:

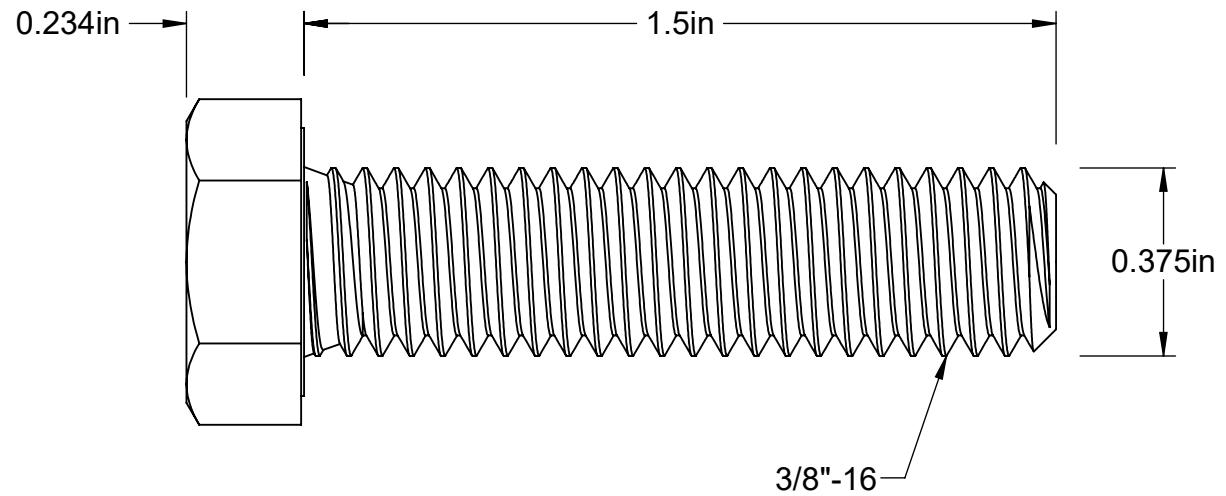
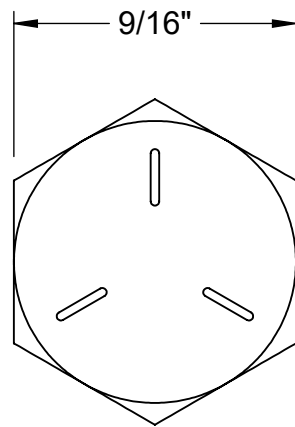
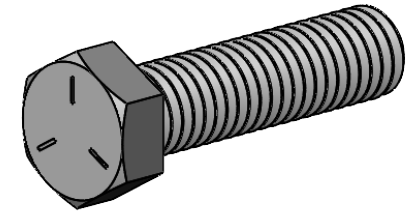
STEEL

PART #:

P-160GI-004

REV:

0



McMASTER CARR
PART NUMBER
92865A628

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm$.01 .XXX $\pm$.005
FRACTIONS $\pm$ 1/64
ANGULAR MACHINED $\pm$.5°
ANGULAR BEND $\pm$ 1°
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET1 OF 1



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

Steel Hex Head Screw

SIZE:

A

DATE:

11/29/2022

DRAWN BY:

LC

MATERIAL:

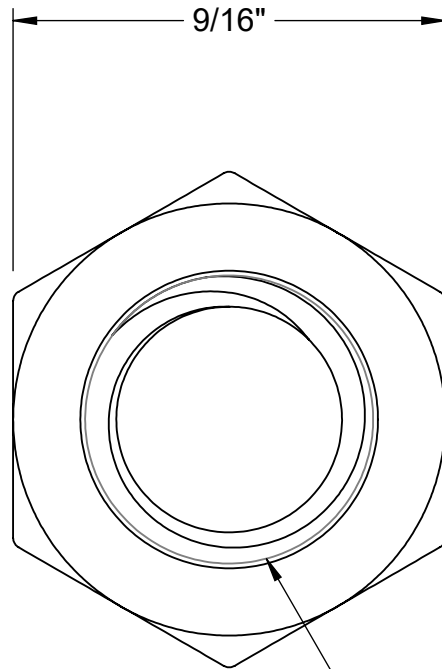
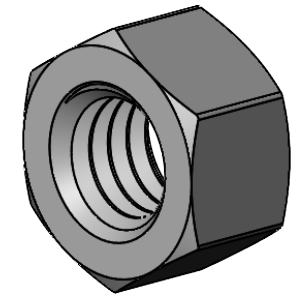
Steel

PART #:

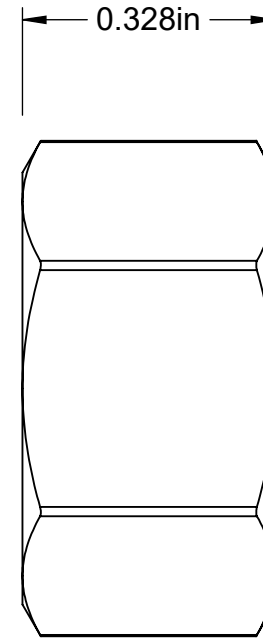
P-160GI-001

REV:

1



3/8"-16



**McMASTER CARR
PART NUMBER
95462A031**

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX ±.01 .XXX ±.005
FRACTIONS ±1/64
ANGULAR MACHINED ±.5°
ANGULAR BEND ±1°
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET1 OF 1



**CASE SCHOOL
OF ENGINEERING**

**CASE WESTERN RESERVE
UNIVERSITY**

TITLE:

Steel Hex Nut

SIZE:

A

DATE:

7/3/2012

DRAWN BY:

LC

MATERIAL:

ZINC-PLATED STEEL

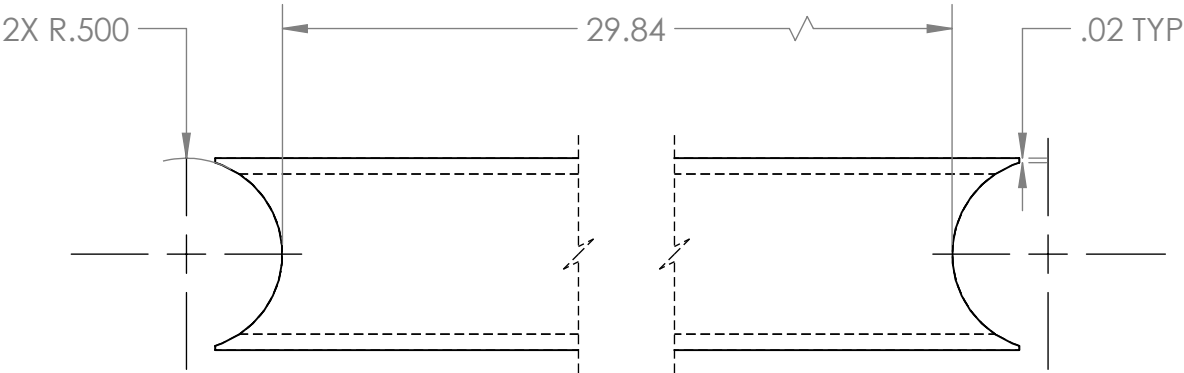
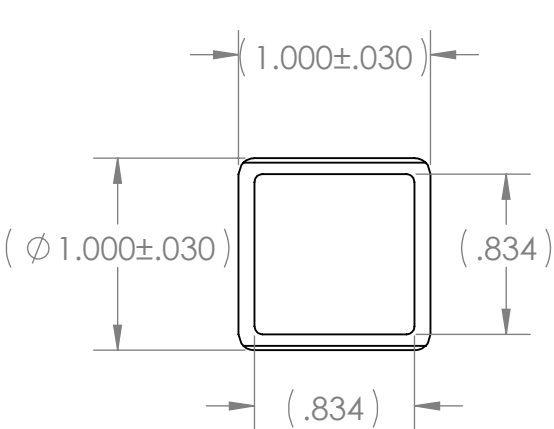
PART #:

P-160GI-002

REV:

1

REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
1	GROUND DOWN SHARP END EDGES	11/18/2025	ID



Made From
McMaster Carr
Part Number
6527K262

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX ±.01 .XXX ±.005
FRACTIONS ±1/64
ANGULAR MACHINED ±.5°
ANGULAR BEND ±1°
SURFACE FINISH
BREAK ALL EDGES .030

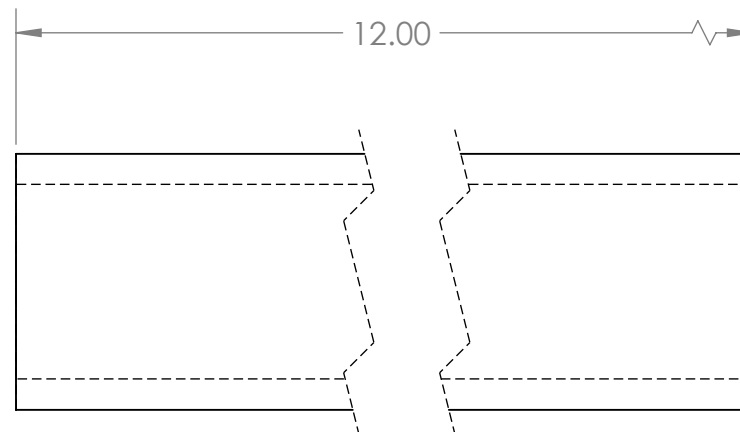
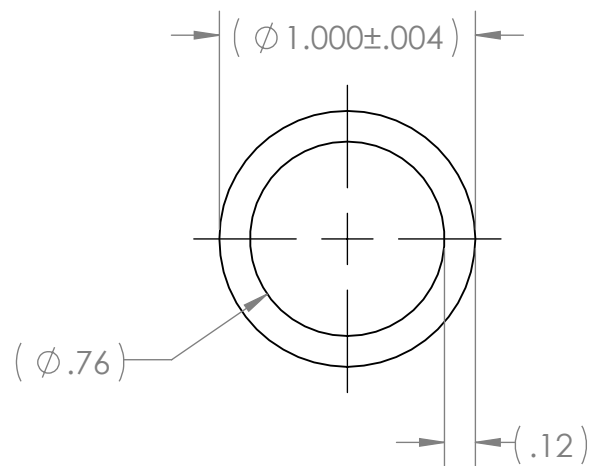


CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE: MAIN SHAFT			
SIZE: A	DATE: 10/30/2025	DRAWN BY: ID	
MATERIAL: Low-Carbon Steel	PART #: M-160GI-007	REV: 1	

DO NOT SCALE
SHEET 1 OF 1



Manufactured From
McMaster-Carr Part
7767T372

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm .01$.XXX $\pm .005$
FRACTIONS $\pm 1/64$
ANGULAR MACHINED $\pm .5^\circ$
ANGULAR BEND $\pm 1^\circ$
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET 1 OF 1



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

HANDLE

SIZE:

A

DATE:

10/27/2025

DRAWN BY:

AXH1066

MATERIAL:

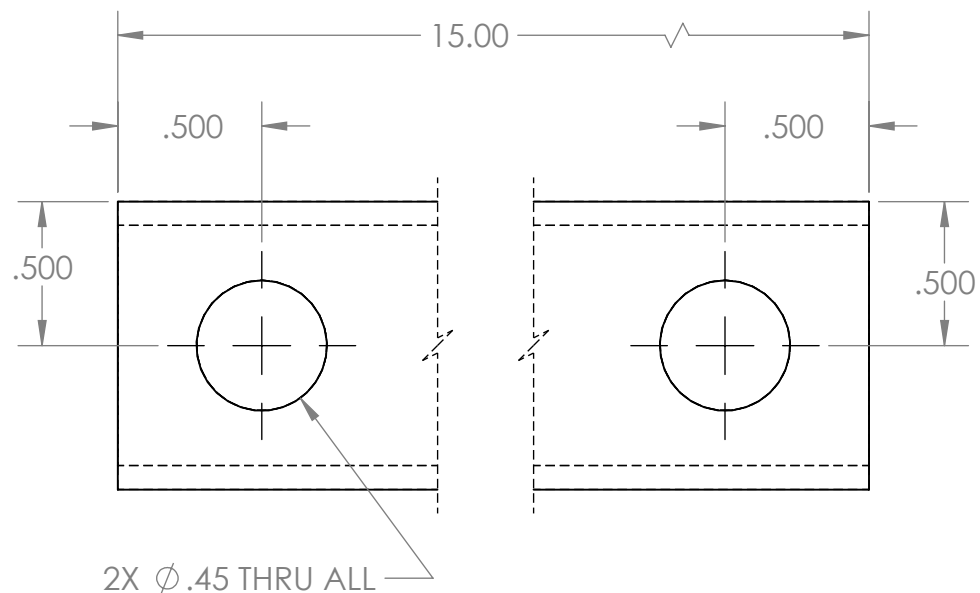
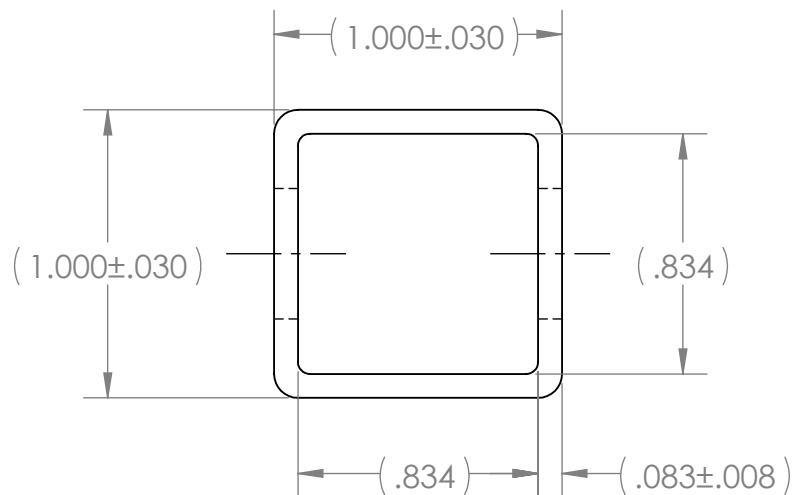
Low-Carbon Steel

PART #:

M-160GI-003

REV:

0



Made From McMaster Carr
Part Number 6527K262

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm .01$.XXX $\pm .005$
FRACTIONS $\pm 1/64$
ANGULAR MACHINED $\pm .5^\circ$
ANGULAR BEND $\pm 1^\circ$
SURFACE FINISH
BREAK ALL EDGES .030



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

COMPRESSION SHAFT

SIZE:

A

DATE:

10/28/2025

DRAWN BY:

ID

MATERIAL:

LOW CARBON STEEL

PART #:

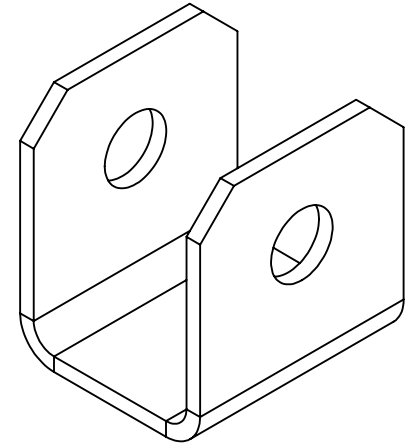
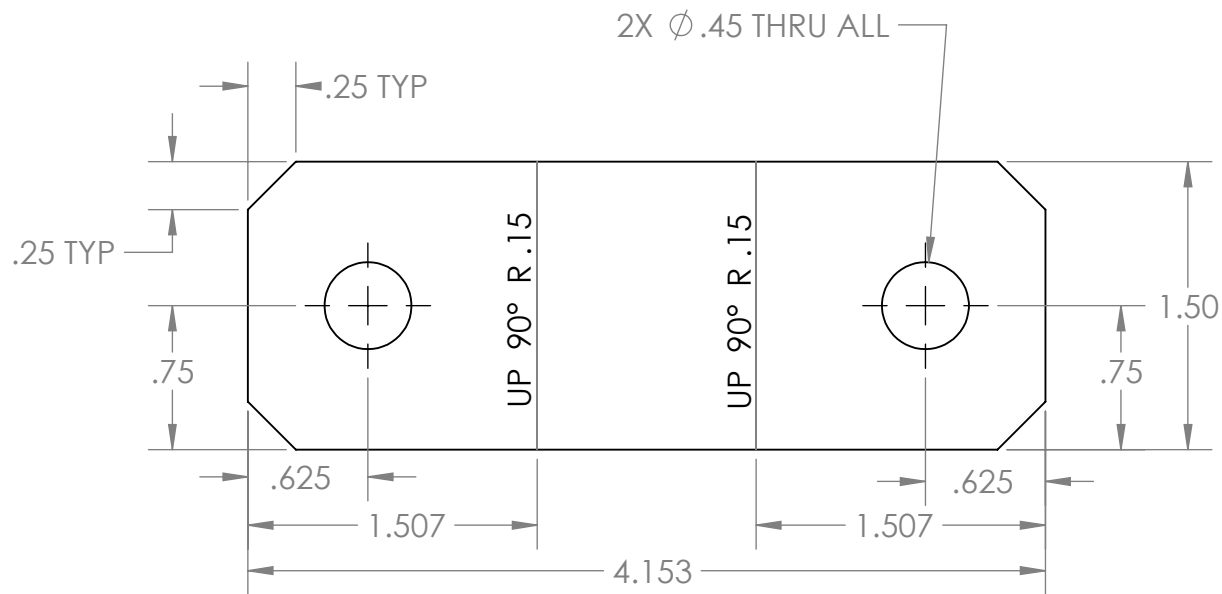
M-160GI-005

REV:

0

DO NOT SCALE

SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm$.01 .XXX $\pm$.005
FRACTIONS $\pm$ 1/64
ANGULAR MACHINED $\pm$.5°
ANGULAR BEND $\pm$ 1°
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET 1 OF 1



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

HINGE BLOCK

SIZE:

A

DATE:

10/29/2025

DRAWN BY:

ID

MATERIAL:

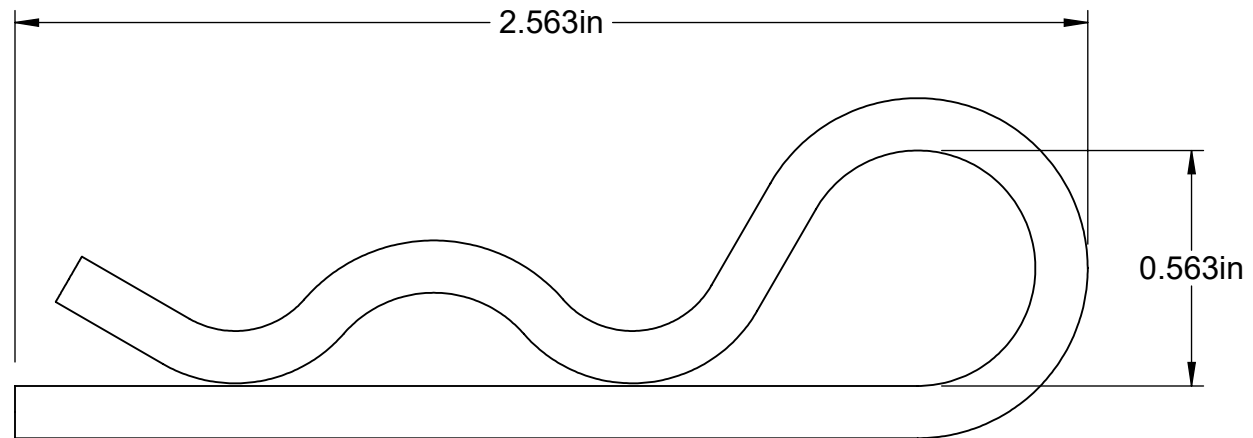
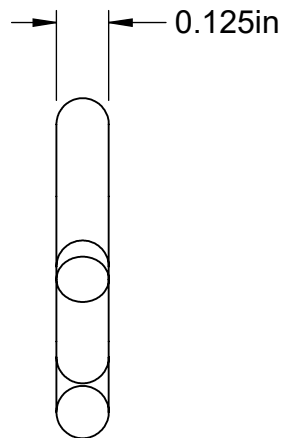
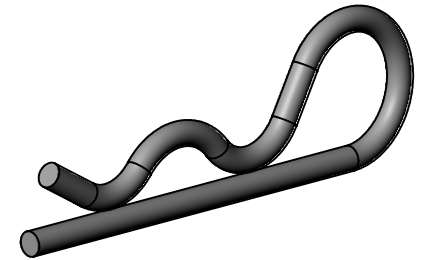
0.1" 4130 Alloy Steel
Sheet

PART #:

M-160GI-002

REV:

1



McMaster Carr
Part Number
98335A127

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm .01$.XXX $\pm .005$
FRACTIONS $\pm 1/64$
ANGULAR MACHINED $\pm .5^\circ$
ANGULAR BEND $\pm 1^\circ$
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET1 OF 1



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

PIN LOCK

SIZE:

A

DATE:

6/29/2021

DRAWN BY:

ID

MATERIAL:

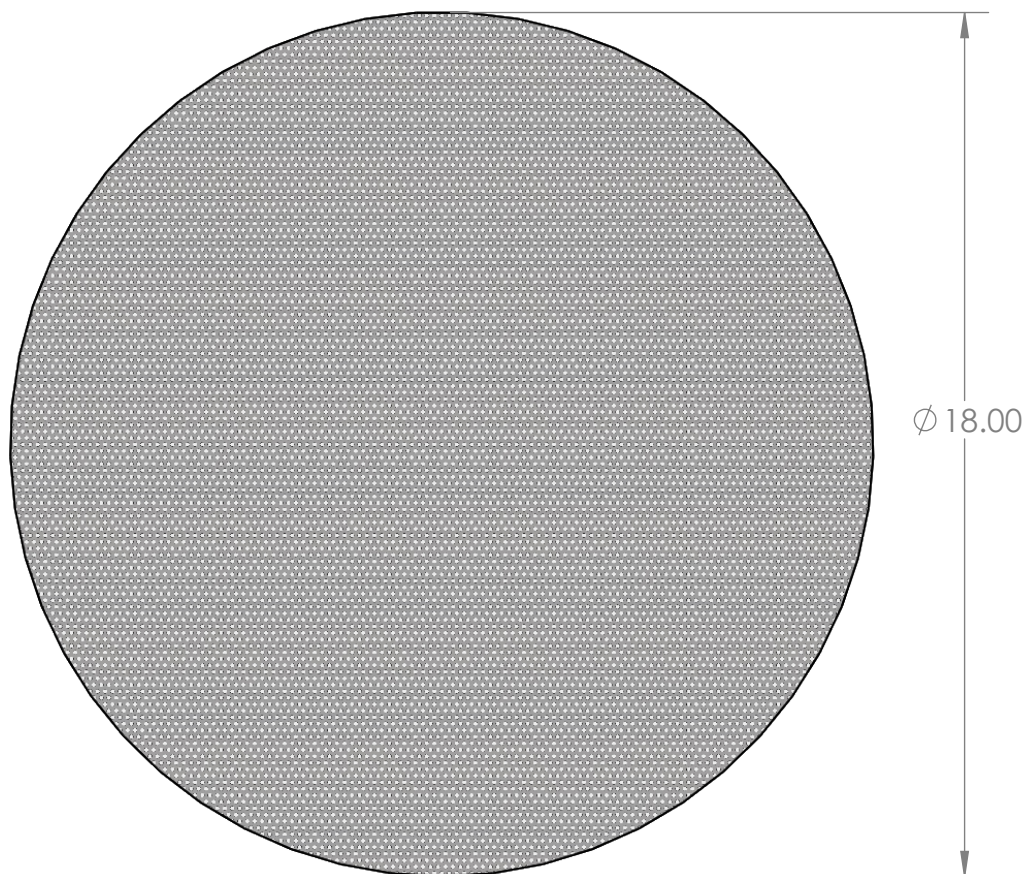
PLATED SPRING STL

PART #:

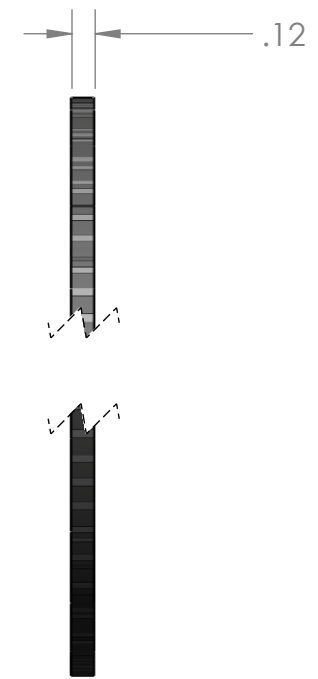
P-160GI-005

REV:

0



REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
1	ADDED SHADING TO SHOW HOLES	11/18/25	NTA25



MANUFACTURED FROM:
MCMASTER CARR PART
NUMBER 9358T561
Perforated Sheet

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX ±.01 .XXX ±.005
FRACTIONS ±1/64
ANGULAR MACHINED ±.5°
ANGULAR BEND ±1°
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET 1 OF 1

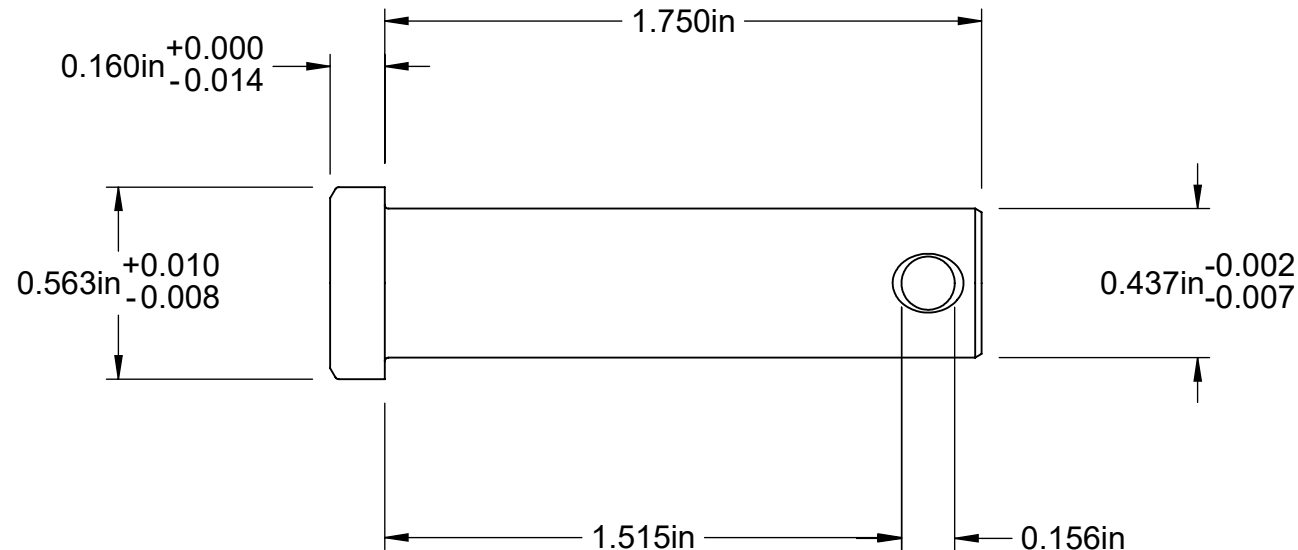
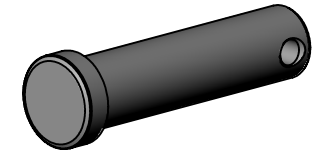


CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE: COMPRESSION PLATE

SIZE: A	DATE: 11/16/2025	DRAWN BY: NTA25	
MATERIAL: 304 Stainless Steel		PART #: M-160GI-001	REV: 1



McMaster Carr
Part Number
98306A330

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
.XX $\pm$.01 .XXX $\pm$.005
FRACTIONS $\pm$ 1/64
ANGULAR MACHINED $\pm$.5°
ANGULAR BEND $\pm$ 1°
SURFACE FINISH
BREAK ALL EDGES .030

DO NOT SCALE

SHEET1 OF 1



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

CLEVIS PIN

SIZE:

A

DATE:

3/14/2023

DRAWN BY:

ID

MATERIAL:

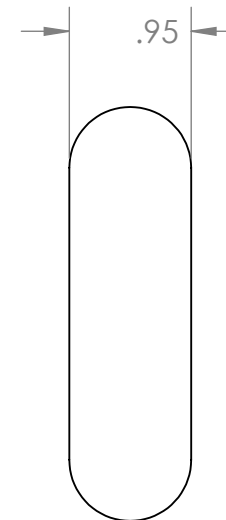
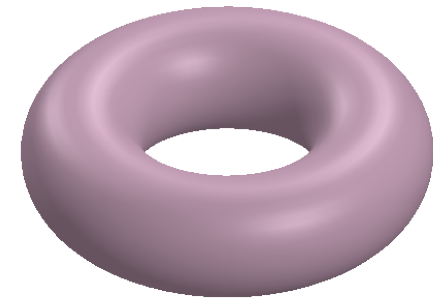
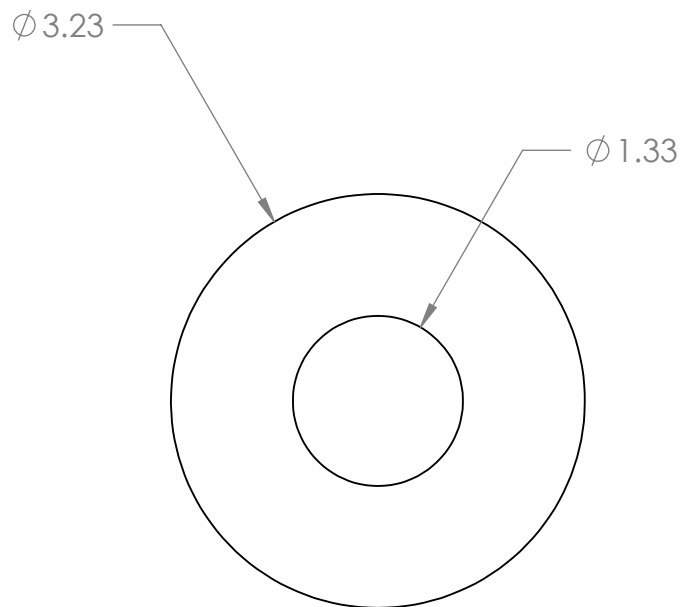
CARBON STEEL

PART #:

P-160GI-003

REV:

0



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS IN INCHES

TOLERANCES:
 $.XX \pm .01$ $.XXX \pm .005$
 FRACTIONS $\pm 1/64$
 ANGULAR MACHINED $\pm .5^\circ$
 ANGULAR BEND $\pm 1^\circ$
 SURFACE FINISH
 BREAK ALL EDGES $.030$

DO NOT SCALE

SHEET 1 OF 1



CASE SCHOOL
OF ENGINEERING

CASE WESTERN RESERVE
UNIVERSITY

TITLE:

Study Tool

SIZE:

A

DATE:

10/30/2025

DRAWN BY:

LC

MATERIAL:

Carbs

PART #:

1

REV:

1